

ORHAN AYDIN

Mechatronics Engineer | AI & Robotics

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UK work authorisation: Student visa valid until 05/10/2026; eligible for the unsponsored Graduate Route until approximately October 2028.

SUMMARY

Mechatronics Engineer and MSc Data Science & AI graduate (Distinction) focused on robotics, embodied AI, and robot manipulation. Designed and built a humanoid robot end-to-end — combining 3D-printed mechanics, embedded control, computer vision, and a custom-trained 1B-parameter language model for natural-language task execution — with deterministic validation preventing unsafe LLM-generated actions. Nearly 2 years of industrial engineering experience delivering measurable production improvements. Skilled in Python, C++, PyTorch, OpenCV, MuJoCo, SolidWorks, and Arduino.

EDUCATION

MSc Data Science and Artificial Intelligence — Distinction 01/2025 – 07/2026

Bournemouth University, United Kingdom — Weighted Average: 75.56%

BSc Mechatronics Engineering 09/2016 – 09/2022

Erciyes University, Turkey

PROJECTS

LLM-Enabled Humanoid Robot (MSc Thesis) 2025 – 2026

orhanaydinn.github.io/projects/humanoid · github.com/orhanaydinn/oax_humanoid_robot

- **Achieved 95% target-object detection and 85% object-centring accuracy** on a physical humanoid robot, by building a custom fixed-base platform with a 3D-printed 5-DOF arm, parallel gripper, 2-DOF pan-tilt head, OAK-D depth camera, and Arduino-based control.
- **Prevented 100% of unsafe LLM-generated executions while maintaining 100% valid JSON outputs**, by designing a deterministic Planner–Controller that validates, repairs, rejects, or clarifies every action before physical execution.
- Enabled natural-language pick-and-place by integrating object detection, depth perception, inverse/forward kinematics, and JSON-based robot tool calling into one end-to-end pipeline; stage-level testing identified grasp stability as the main physical limitation.

Robot Learning for 5-DOF Arm Manipulation — MuJoCo 2026

github.com/orhanaydinn

- **Achieved a 96% simulated pick-task success rate** across varying table heights, using Behaviour Cloning on structured demonstrations and displacement-aware reward shaping during Reinforcement Learning fine-tuning.
- Implemented action chunking so the policy predicts short future action trajectories instead of single-step commands, improving temporal consistency and reducing unstable joint motion.
- Built a reproducible BC-to-RL pipeline for reach, approach, grasp, and lift, with a custom MuJoCo environment, automated demonstration collection, classical skill sequencing, and metric-based comparison of BC-only and RL-fine-tuned policies.

OAX-1B: Custom LLM for Robot Tool Calling 2025 – 2026

huggingface.co/orhanaydinn/OAX-1B-Humanoid

- Delivered the reasoning layer of the humanoid robot, as measured by **100% JSON schema validity** and **77.8% raw tool-call accuracy**, by pre-training a 1B-parameter LLaMA-style model from scratch (custom tokenizer, checkpointed training) and supervised fine-tuning it with LoRA on custom human–robot interaction data.
- Constrained model outputs to a strict four-field JSON schema with chat / tool_call / clarify / refuse behaviours, by designing the fine-tuning dataset and prompt format around six high-level robot tools (search, pick, place, status, visibility, stop).

SCARA Robot — Mechanical Design, Manufacturing & Control 2022

github.com/orhanaydinn/scara-robot-design-and-control

- Designed and built a working SCARA robotic arm from scratch, including the SolidWorks mechanical assembly, 3D-printed components, actuator integration, and end-effector structure.
- Developed Arduino and CNC-shield motion control with a Processing-based operator interface, applying forward/inverse kinematics, calibration, and iterative testing for coordinated and repeatable joint motion.

PROFESSIONAL EXPERIENCE

Mold Specialist Engineer — Yorglass

08/2023 – 08/2024

Turkey · Industrial glass production

- **Increased production speed by 27% (10 to 12.7 rpm)**, by redesigning glass press machine molds in SolidWorks and AutoCAD and optimising the pressing process for automated lines.
- **Reduced robot-related production losses from 25% to 5% per day**, by designing customised robotic gripper solutions and collaborating with production and maintenance teams on handling reliability.

Maintenance Engineer — Yorglass

10/2022 – 08/2023

Turkey

- Improved handling and production flow on automated glass press lines, by implementing image-processing-based robot transfer solutions and performing mechanical and electrical maintenance on industrial systems.

Voluntary Project Engineer — MEM R&D

10/2019 – 05/2022

Turkey

- Guided student teams to national competition readiness (Teknofest, TUBITAK), by mentoring embedded-system design, prototyping, project planning, and technical presentations.

SKILLS

Robotics & Simulation: Robot Manipulation, Robot Learning, Behaviour Cloning, Reinforcement Learning, MuJoCo, Inverse/Forward Kinematics, Motion Planning, Sim-to-Real, Grippers, Sensor Integration

Computer Vision & Embodied AI: OpenCV, YOLO, Object Detection, Depth Perception, Visual Servoing, Vision-Language Robot Control, LLM Tool Calling, Human-Robot Interaction

AI & Machine Learning: PyTorch, TensorFlow, Keras, Transformers, Hugging Face, PEFT/LoRA, Deep Learning, Large Language Models

Programming & Tools: Python, C++, SQL, Git, Docker, Streamlit

Embedded & Hardware: Arduino, STM32, ESP32, Raspberry Pi, Embedded Control, Electronics Integration

CAD & Mechanical Design: SolidWorks, AutoCAD, Altium Designer, 3D Printing, Mechanical Design, Prototyping

CERTIFICATIONS (SELECTED)

- Deep Learning with PyTorch and Keras — BTK Akademi (2025)
- Machine Learning Applications with Python — BTK Akademi (2025)
- Python & OpenCV for Image Processing — Udemy (2025)
- PCB Design with Altium Designer — Udemy (2022)

LANGUAGES

English — professional working proficiency (MSc completed in English, UK) · Turkish — native